

Shebar Janala

A benefits-navigation and local-accountability system, and an honest account of what it does not yet do

1. The problem, and the position this system takes

A woman in Rangpur whose husband has just died is entitled to a monthly allowance. Whether she receives it depends on facts that have nothing to do with her entitlement: whether she knows the programme exists, whether anyone can tell her if she qualifies, and whether the person she asks happens to be right. If the answer she gets is wrong, the cost is not an error message. It is a day's wage and a bus fare to a district office, or years of an allowance never claimed.

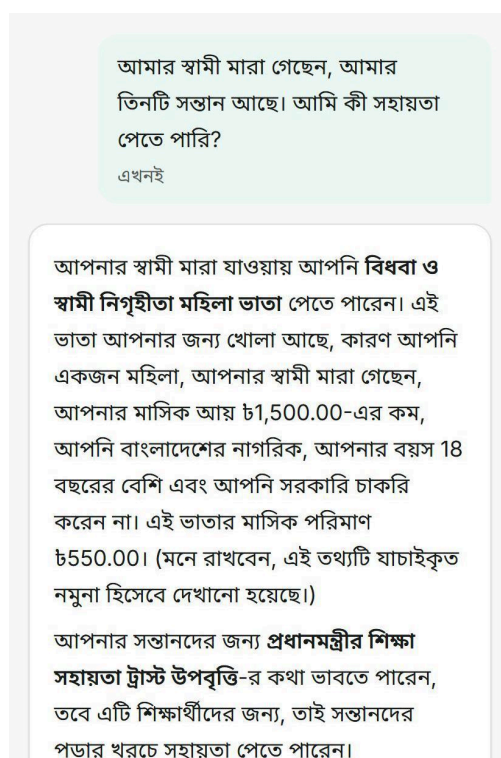


Figure 1. The running prototype answering that exact question in Bangla. The citizen describes a life event; the system names the Widow and Husband-Deserted Women Allowance and states a monthly amount.

The design position

Given that the expensive failure is confident wrongness, the system is organised around making wrongness detectable rather than merely unlikely. Three commitments follow, and they are enforced in types and data rather than in policy documents.

First, the eligibility verdict is a pure function of a stored profile snapshot and a versioned rule, so any decision can be recomputed months later and compared against what the citizen was told. Second, not-knowing is a first-class value: unknown is one of four possible eligibility outcomes, a missing field never silently becomes a no, a ranking factor with no data behind it reports itself unavailable rather than guessing, and a simulated identity check is stored under the literal status simulated_verified.

Third, the money layer is hash-chained, so an edit to a historical financial record is detectable by re-running a script that anyone with database access can run outside the application.

2. Architecture and trust boundaries

One Next.js 15 application (React 19, TypeScript strict) over Drizzle ORM and libSQL, organised as 19 modules behind a single deployment boundary. It runs with no external database server, no cache, and no API key. Every external dependency is optional and key-gated; with none configured the product still answers citizens, using lexical search and a deterministic text composer.

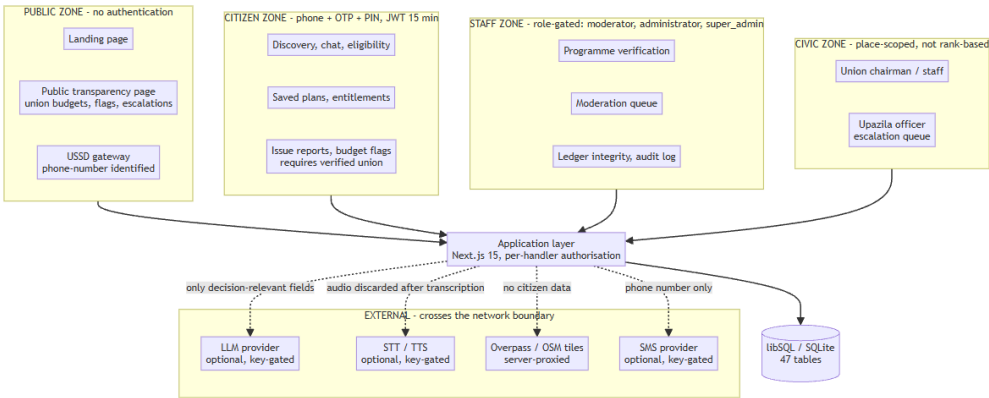


Figure 2. Trust zones and the network boundary. Note what leaves the system: the LLM receives only decision-relevant profile fields, the speech provider receives audio that is discarded after transcription, the map proxy sends no citizen data, and the SMS provider receives only a phone number.

Verified implementation figures

Metric	Whitepaper	Verified	Method
Database tables	32	47	sqliteTable definitions in src/lib/db/schema.ts
API route files	31	58	route.ts files under src/app/api/v1
Exported HTTP handlers	51	84	Exported GET/POST/PUT/PATCH/DELETE
Automated tests	685	789 passing	npm test: 26 files, 789 tests, 7.17 s
Module domains	19	19	Directories under src/modules
Programme records	42	42	Seed corpus, all unverified_sample
Retrieval chunks	158	158	Deterministic: 34 records to 4 chunks, 6 to 3, 2 to 2

Deployment, current and target

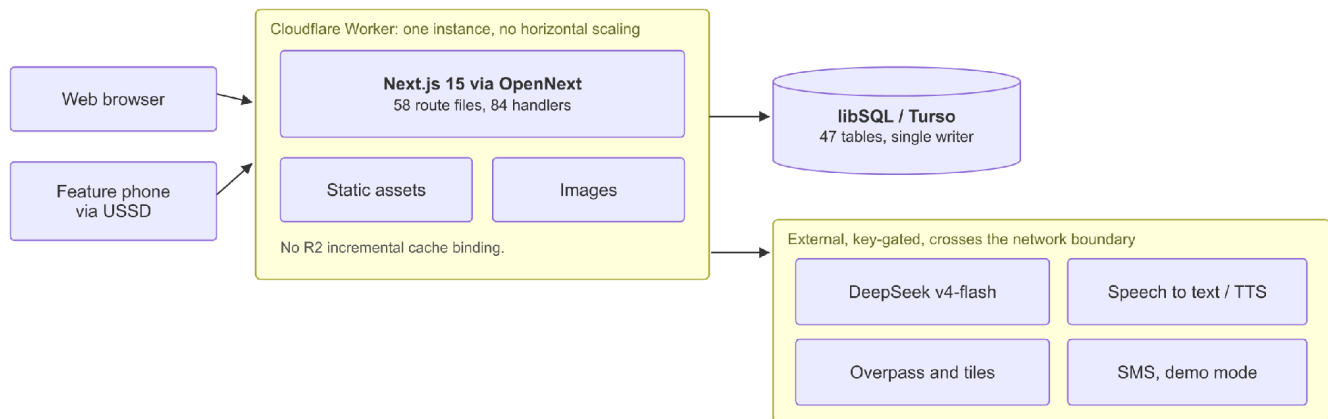


Figure 3. Runtime and deployment topology: one Worker instance, one durable store, and four optional externals, all key-gated. There is no OpenNext R2 cache binding now, so Turso holds everything that outlives a request. Four gaps in this deployment: no scheduled trigger (O-01), no backup or restore (O-02), a schema that is pushed rather than migrated (O-03), and a ledger append that assumes the single writer (G-04).

Concern	Today (IMPLEMENTED)	Target (SPECIFIED)
Runtime	Single Next.js instance on Cloudflare Workers via OpenNext	Two or more stateless instances behind managed load balancing
Database	libSQL / SQLite, single writer	PostgreSQL with pgvector for the embedding channel
Ledger appends	Read-then-write, protected by a uniqueness constraint (Section 6)	Serialised appends: advisory lock or single-writer queue, required before horizontal scaling
Inference	DeepSeek deepseek-v4-flash, named in wrangler.jsonc and gated on DEEPSEEK_API_KEY. Other providers remain optional third-party HTTP APIs	In-country host for speech and extraction where data residency is required
Background jobs	Six jobs, manual HTTP trigger only, run inline in the request	Scheduled runner with retry and a dead-letter path (GAP O-01)
Backups	None (GAP O-02)	Automated snapshot with a stated RPO and a tested restore
Caching	None. libSQL and Turso hold everything durable; the OpenNext R2 cache binding was dropped from wrangler.jsonc and open-next.config.ts to keep the deployment card-free	Re-introduce R2 or an equivalent edge cache when traffic justifies it

3. Data model, and the reconciliation gap

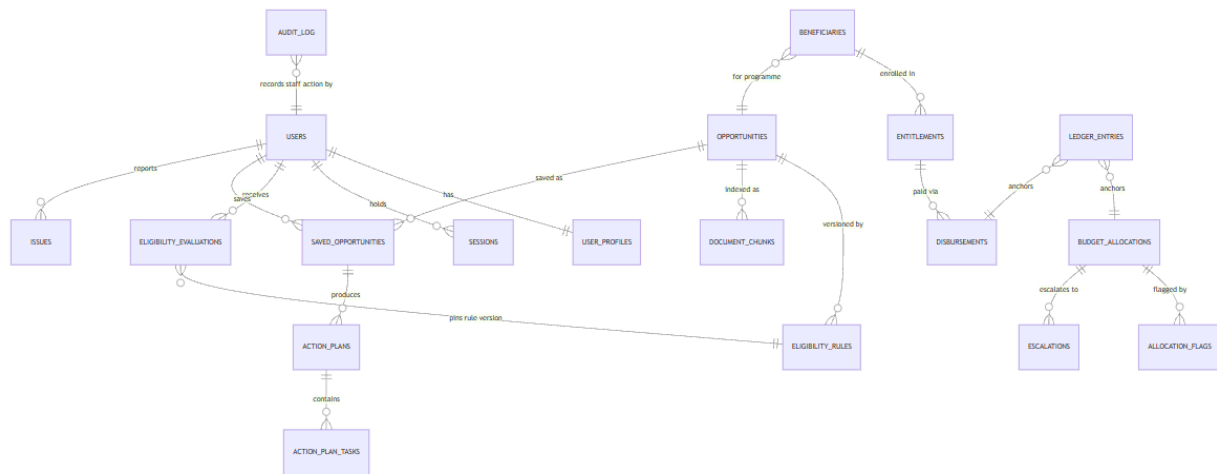


Figure 4. Core entities. The left cluster is what the system calculates for a citizen; the right cluster is what a union actually enrolled and paid. They are joined by nothing.

Two truths that never meet

The system maintains two independent records of a citizen's relationship to a programme. Eligibility evaluations record what the rules say a person should qualify for. Beneficiary, entitlement, and disbursement records record what a union actually enrolled and paid. These are correctly modelled as separate concerns, because a calculation is not an enrolment. But nothing reconciles them.

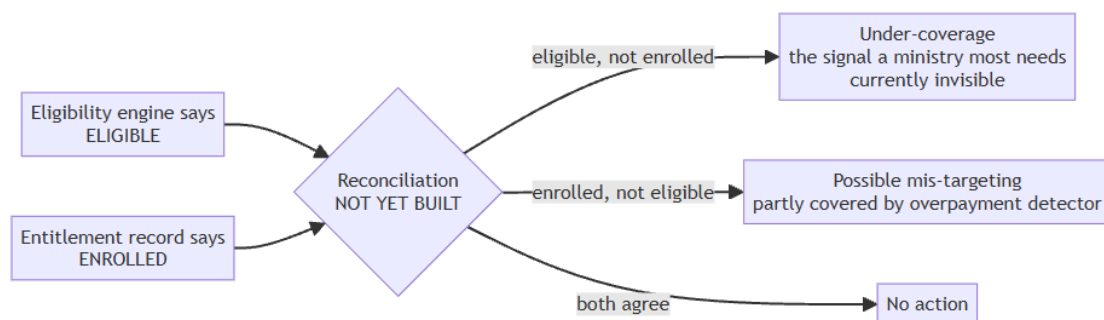


Figure 5. The reconciliation that does not exist. Five oversight detectors examine records that already exist, so the system can see money going to the wrong place and cannot see money never reaching the right one.

This matters more than any other data-model observation in this document. The five oversight detectors described in Section 6 all read tables of things that happened: allocations, beneficiaries, disbursements, escalations. Not one of them asks whether a citizen the engine found eligible was ever enrolled. The system therefore detects leakage and is blind to under-coverage, which is the failure mode the National Social Security Strategy names as the harder problem, and the one a ministry would most want a tool to surface. It is issue D-01 in Section 13.

Replayability and its limit

Every evaluation stores the citizen's full profile snapshot and the rule version applied, so a verdict can be recomputed exactly. That guarantee holds only while the meaning of those stored structures is stable: rename a profile field or change a comparator's semantics and an old snapshot replays differently or not at all. No schema version is stored alongside the snapshot. Issue D-02, low severity today because the schema has not yet changed under a live corpus, and cheap to fix now by stamping a schema version on write.

Concurrency

The codebase contains no database transactions. A repository-wide search for transaction and batch calls returns nothing. Two consequences are described where they bite: the verify-and-edit race in Section 4 and the ledger append race in Section 6. Both are issue D-03.

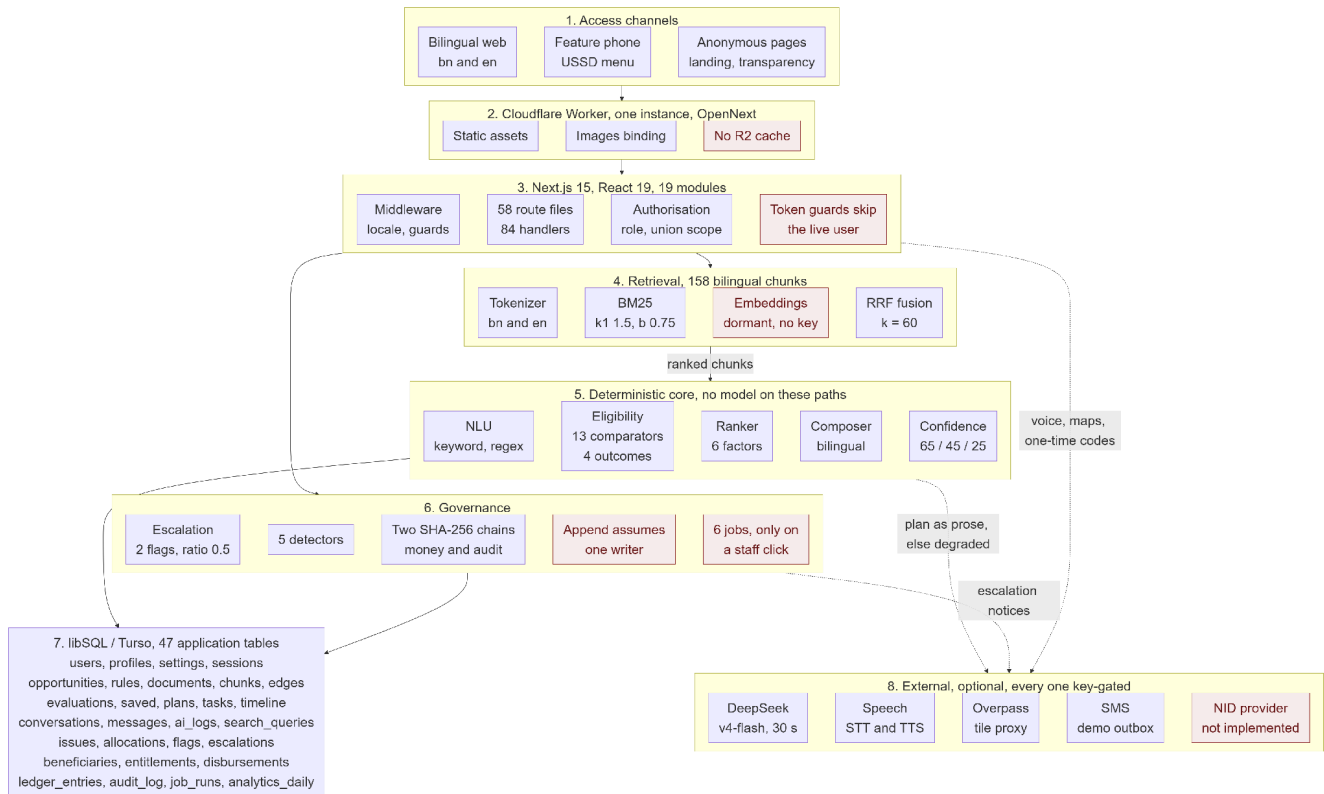


Figure. The whole system on one page, layer by layer. The counts are those re-counted from the working tree: 58 route files and 84 handlers across 19 modules, 158 retrieval chunks, 47 application tables. Red boxes are gaps this document already carries an issue for, namely the removed R2 cache binding, token guards that skip the live user, the dormant embedding channel, a chain append that assumes one writer, jobs that run only when staff click, and the NID provider that is deliberately unimplemented.

4. The decision layer: deterministic eligibility

The engine evaluates a validated JSON rule structure over 13 comparators against a profile: eq, neq, gt, gte, lt, lte, in, not_in, between, exists, not_exists, contains_any, contains_all. Conditions are hard (a statutory bar decides immediately) or soft (weighted, contributing to a score). No language model can reach this code path.

```
case 'gt':
case 'gte':
case 'lt':
case 'lte': {
  const a = typeof actual === 'number' ? actual : Number(actual);
  ...
  if (condition.operator === 'gt') return a > e ? 'met' : 'failed';
  if (condition.operator === 'gte') return a >= e ? 'met' : 'failed';
  if (condition.operator === 'lt') return a < e ? 'met' : 'failed';
  return a <= e ? 'met' : 'failed';
}

// src/modules/eligibility/engine.ts
```

The outcome is always one of `eligible`, `partially_eligible`, `not_eligible`, or `unknown`. A field the citizen has not supplied yields `unknown` and never a negative verdict. This is the difference between a system that says no and a system that says it does not yet know, and for a benefits product that distinction is the whole ethical content of the design.

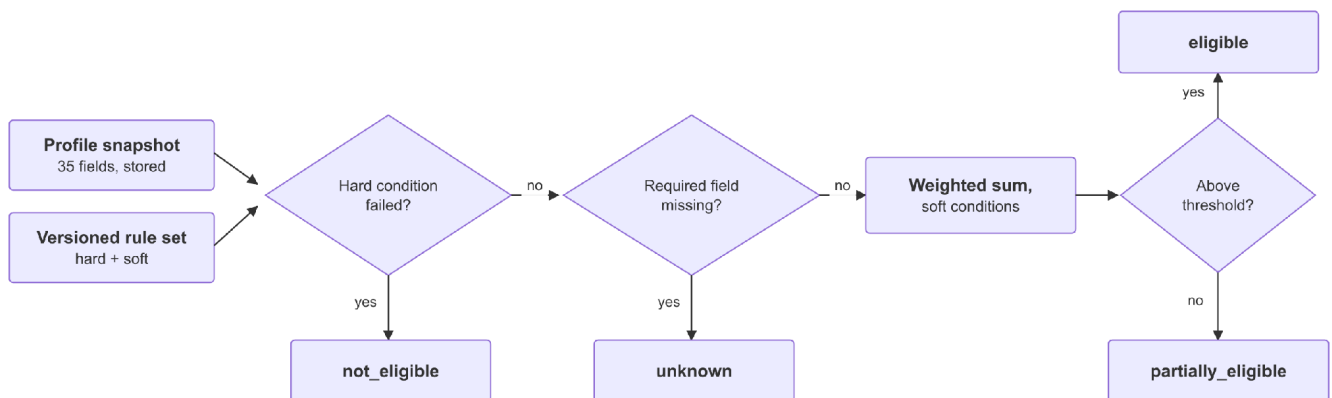


Figure 6. The decision path. A hard condition decides immediately, a field the citizen never supplied yields `unknown` rather than a negative verdict, and only soft conditions reach the weighted score. No language model can reach this path; the verdict is a pure function of the stored profile snapshot and the rule version, so it can be recomputed months later and checked against what the citizen was told.

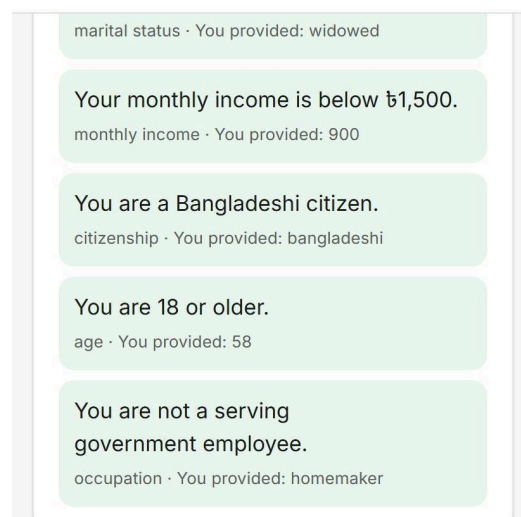


Figure 7. The per-condition trace a citizen sees. Each line names the rule, the profile field, and the value the citizen supplied, so a caseworker can audit the verdict without database access.

Verification cannot be edited into existence

Verifying a programme record and editing its content cannot happen in one request; the handler rejects a PATCH that attempts both. Editing a substantive field on a verified record forces the record back to pending review and clears its last-verified date. The race remains: two administrators acting simultaneously, one verifying while the other edits, are not serialised, because the codebase has no transactions. Specified fix in Appendix A, item 3.

Ranking never overrides the verdict

Ordering within a result set is a weighted sum: eligibility fit 40 percent, stated preference 15, location 15, deadline urgency 10, popularity 10, and a similar-user-success factor of 10 that is hardcoded to a neutral value and reports `similarUserDataAvailable` false to the client, because there is no outcome

data to learn from. Ranking never lifts an ineligible programme above an eligible one; the weights order results inside an outcome band only.

5. The AI layer, with measured retrieval results



Figure 8. One citizen turn. Retrieval and the eligibility engine complete before any model is invoked, and produce the same facts whether or not a provider is configured.

5.1 Retrieval, and what measuring it revealed

Search runs Okapi BM25 ($k_1 = 1.5$, $b = 0.75$, IDF smoothing) over 158 bilingual chunks, with hand-built tokenisation: Unicode-aware Bangla segmentation, a curated Bangla suffix stemmer, a Porter-style English stemmer, and separate stopword lists. A semantic channel embeds the query through a configured provider and is combined with the lexical ranking by Reciprocal Rank Fusion at $k = 60$, chosen over a weighted score average because BM25 scores and cosine similarity are not on comparable scales. Without an embedding key that channel returns nothing and contributes nothing.

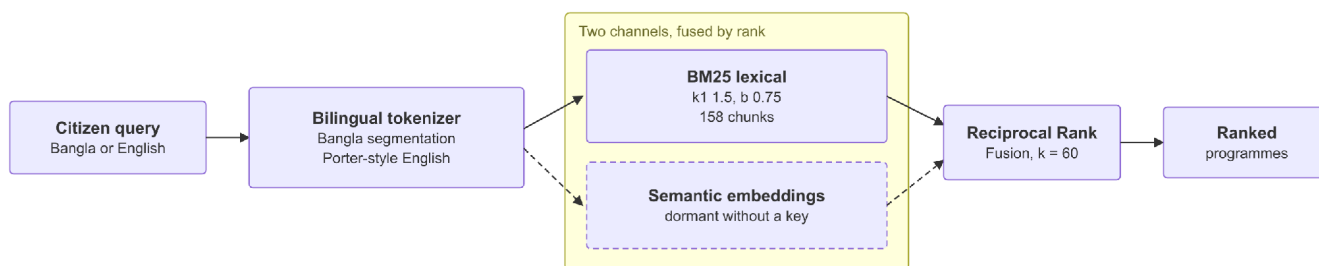


Figure 9. The retrieval pipeline. The semantic channel is dashed because it is dormant without a provider key, so every number here comes from the lexical channel alone: Recall@1 85%, Recall@3 90%, MRR 0.867 over 20 labelled known-item queries. English scored 10 out of 10 and Bangla 8 out of 10. Recall is flat from $k=3$ to $k=10$, so both Bangla failures are total misses rather than ranking problems, and the cause is morphological.

A relevance harness covers 20 known-item queries, half Bangla and half English, each labelled with the programme records a caseworker would accept as a correct answer. It runs against the shipped retrieval function, not a reimplement.

```
$ npx tsx scripts/retrieval-eval.ts
```

Retrieval evaluation – 20 labelled queries (10 Bangla, 10 English)
Channel: BM25 lexical only unless an embedding key is configured.

```
Recall@1  17/20  85.0%
Recall@3  18/20  90.0%
Recall@5  18/20  90.0%
Recall@10 18/20  90.0%
MRR      0.867
```

```
Recall@3 by language:  Bangla 8/10  English 10/10
```

Both failures were diagnosed rather than merely counted. A query meaning "I am 70 years old and poor" fails to reach the Old Age Allowance because the corpus writes বয়স্ক (elderly) where the query

says বয়স (age), and writes "no income or very little" where the query says গরিব (poor). A query meaning "a stipend for my children's studies" fails because the corpus writes উপবৃত্তি (stipend) where the query says বৃত্তি (scholarship). That second case is structural: the difference is a prefix, and a suffix stemmer cannot bridge a prefix by construction.

This is the empirical case for the semantic channel. Dense retrieval addresses exactly this class of failure, morphological and synonym mismatch in a low-resource language, and the harness provides the baseline that any future claim of improvement has to beat. Enabling embeddings without re-running it would be a claim with no evidence behind it.

Limits of the measurement: 20 queries is small, the labels were created in-house rather than independently adjudicated, and this measures retrieval mechanics only. It says nothing about whether the retrieved record contents are factually correct, which is issue R-01 and cannot be measured until the corpus is verified.

5.2 Grounded generation, and what does not check it

A deterministic layer assembles a fixed response plan (cited facts, verdict, next steps) before any model is called. A configured model then renders that plan as prose at temperature 0.3 with a 900-token ceiling. Context is deliberately narrow: only seven profile fields reach the prompt (age, gender, district, occupation, monthly income, marital status, education), conversation history is truncated to the last four messages at 400 characters each, and citations are capped at six with 320-character excerpts. Medical conditions, pregnancy, and disability status are never sent to a third-party model.

wrangler.jsonc names the provider for the deployed Worker: DeepSeek deepseek-v4-flash, with chain of thought disabled. The call still needs DEEPSEEK_API_KEY as a Worker secret, and without that key the turn degrades to the deterministic composer described below.

If no provider is configured or a call fails, a deterministic composer renders the same facts, the response is marked degraded, and the citizen is shown that badge. This fallback is the reason the model is not load-bearing.

5.3 Cost, latency, and timeout behaviour

Property	Value	Note
Model call ceiling	900 output tokens, temperature 0.3	Set per request
Provider timeout	30 s per attempt, 1 retry, 400 ms backoff	No jitter
Route abort	45 s, returns HTTP 504	Two provider attempts can exceed this: a design inconsistency, issue A-04
Response caching	None	Every turn is a fresh call
Provider failover	None	One provider is selected from config and cached for the process lifetime; a dead provider degrades to the composer rather than trying another

Property	Value	Note
Deterministic path cost	Zero external calls	Retrieval, eligibility, ranking, and the composer never leave the process

Unit inference cost is deliberately not stated as a figure. Token volume per turn is bounded by the settings above, but the price of those tokens depends on a vendor contract that does not exist yet, and a number invented now would be the kind of unsupported claim the rest of this document avoids. What can be said is the shape: roughly the majority of turns reach no model at all, and the deterministic path has no marginal cost.

5.4 Prompt injection

Retrieved programme text enters the model's context, and under the roadmap that text will eventually be machine-extracted from government circulars. That is a live indirect-injection surface. Today the exposure is bounded because the corpus is authored in-repository rather than fetched, and because the model cannot take actions: it renders text and nothing else, with no tool access, no write path, and no ability to change a verdict. Before any automated ingestion ships, retrieved content must be delimited and marked as data in the prompt, and extraction must be schema-constrained. Issue A-05, currently low exposure and rising the moment ingestion is built.

5.5 What has no model behind it

Life-event detection, intent classification, and entity extraction are keyword lexicons and conservative regular expressions, not trained models, and this document does not present them as AI. Three genuinely model-backed components are proposed and unbuilt: Bangla speech adaptation, learned field extraction, and circular-to-rule document intelligence. Each has a release floor it must clear before it may touch a citizen's profile: 0.98 precision on income, age, and land fields with mandatory abstention below that; word and character error rates reported per district rather than as one average; and exact field match against a human-authored answer key with a mandatory human review step. No training data has been collected for any of them.

6. The governance layer: budget, escalation, ledger

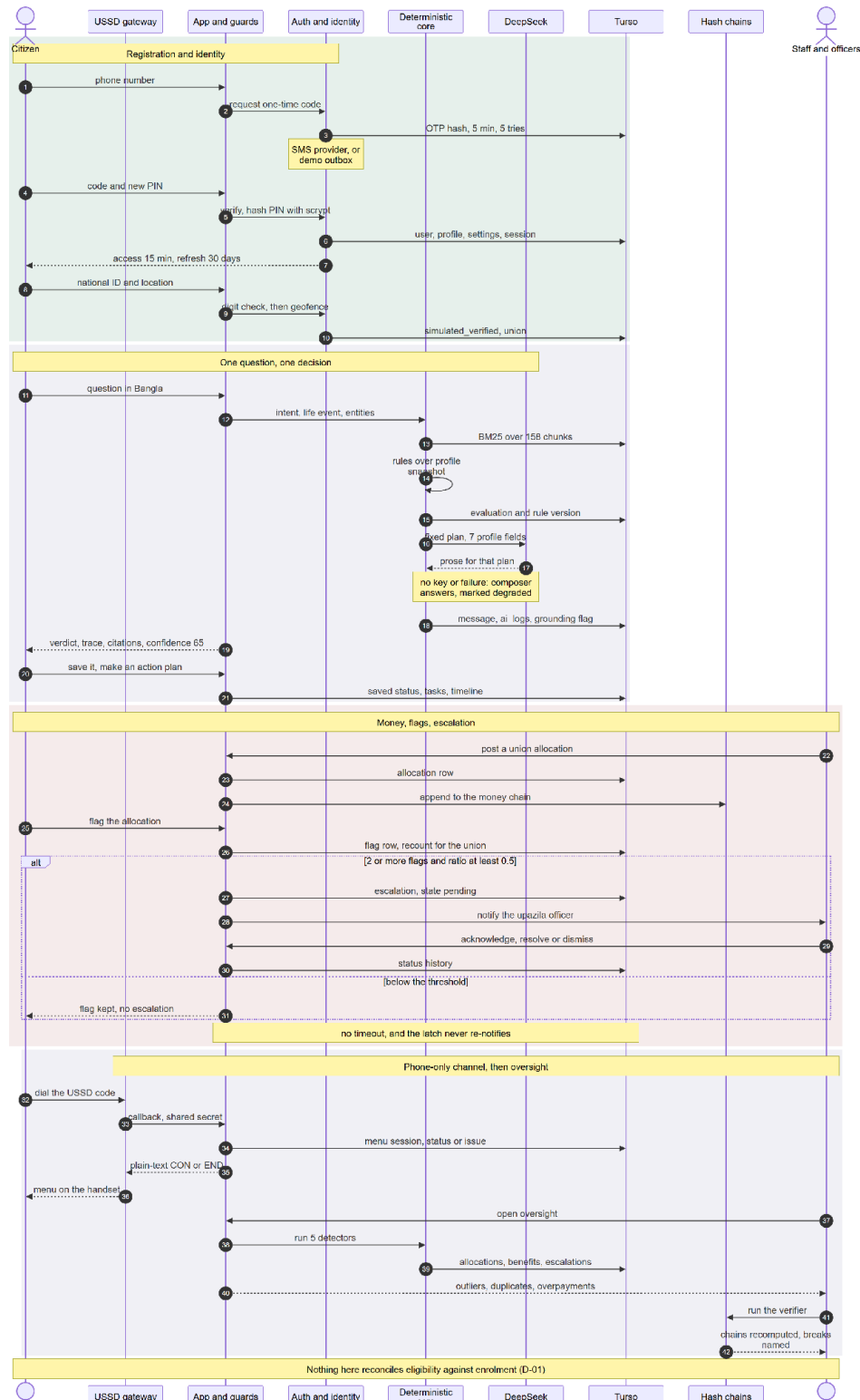


Figure. One pass through the system with every actor present: the citizen, the USSD gateway, the application and its guards, auth and identity, the deterministic core, DeepSeek, the database, the two hash chains, and staff. The timings and thresholds are the values the code enforces. The closing note records what this sequence cannot do, which is reconcile eligibility against enrolment (D-01).

6.1 Flagging and escalation, with the real thresholds

A union allocation is posted and anchored to the ledger. Citizens can flag it. Escalation fires when at least 2 flags exist and the flag count divided by the number of registered residents of that union reaches 0.5.

```
export const ESCALATION_THRESHOLD_RATIO = 0.5;
export const ESCALATION_MIN_FLAGS = 2;

// src/modules/budget/escalation-rules.ts
```

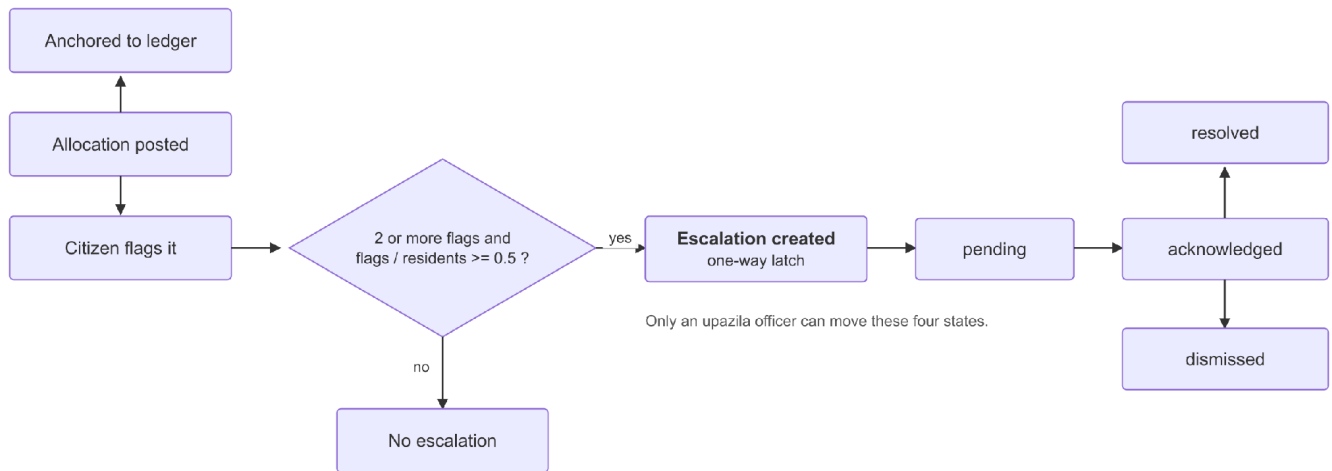


Figure 10. What happens to a flagged allocation, and the three places the chain breaks. The thresholds are the constants shown above. The denominator counts residents whose residency was never verified (G-01), the latch never re-notifies anyone as the flag count keeps rising (G-02), and nothing moves an escalation an officer ignores: no timeout, no service-level target, no automatic promotion, no reassignment (G-03).

Union budget

Emergency Flood Relief Fund
₹50,00,000.00

A single large emergency allocation following unseasonal flooding — deliberately unusual in size.

Posted by মোঃ জসিম উদ্দিন · Kaligonj Union · 19 August 2026

Reason (optional)

⚠ Flag as suspect

Ledger

1 entry — Unaltered since it was recorded

Figure 11. A seeded allocation of 50 lakh taka, deliberately sized to be anomalous, with the citizen flag control and the ledger status shown on the same screen.

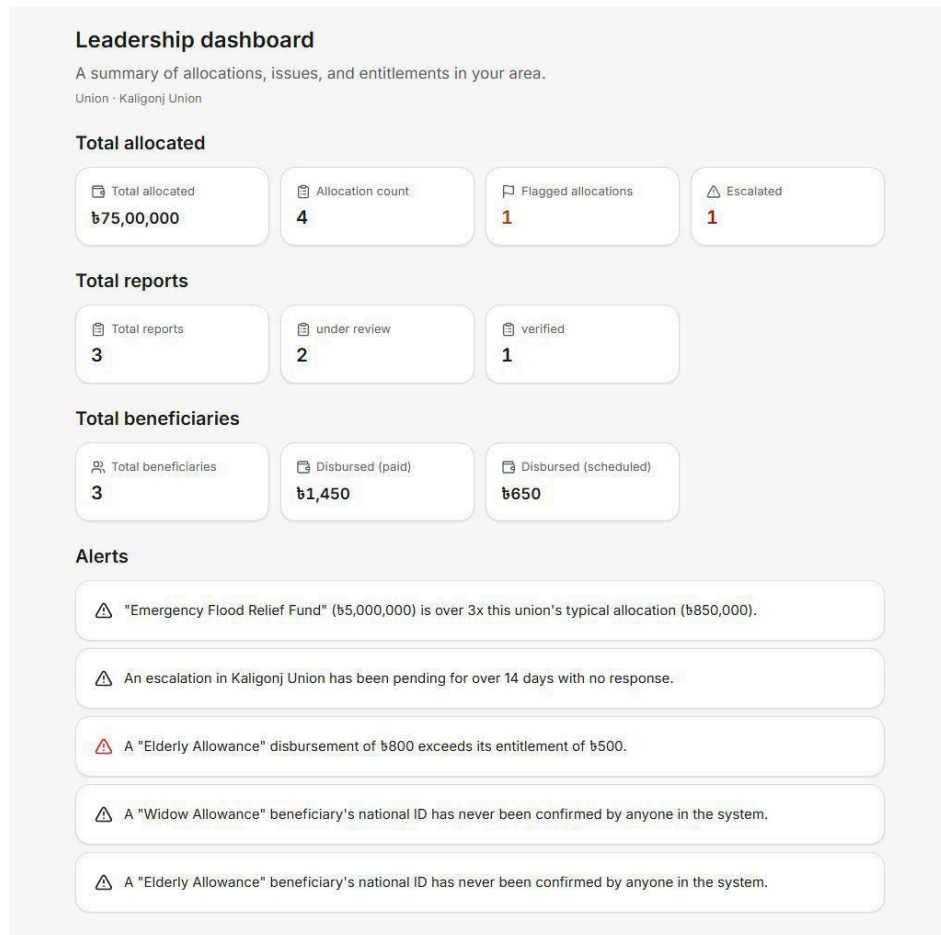


Figure 12. The oversight dashboard naming specific anomalies: an allocation over three times the union's typical size, an escalation pending beyond 14 days, a disbursement exceeding its entitlement, and beneficiaries whose national ID has never been confirmed.

6.2 The five detectors, and the sixth that does not exist

Detector	Rule	Severity
Allocation outlier	Amount over 3x the median of other allocations in the same union, requiring at least 2 comparisons	Medium
Duplicate enrolment	Same national-ID hash and programme active in more than one union	High
Unverified beneficiary identity	Active beneficiary whose ID hash appears in no verified set	Medium
Overpaid disbursement	Disbursement amount greater than the entitlement amount	High
Stale escalation	Status still pending more than 14 days after creation	Medium
Under-coverage: eligible but never enrolled	DOES NOT EXIST. See Section 3 and issue D-01.	Not implemented

6.3 The ledger, and precisely what it proves

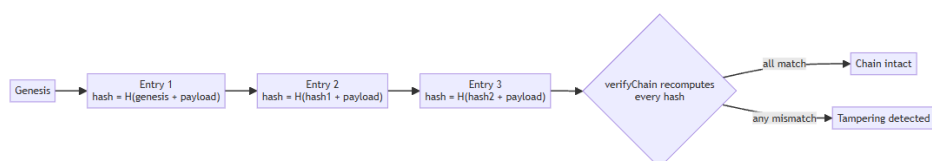


Figure 13. Each entry's hash derives from the entry before it, so an altered or removed row breaks every hash after it.

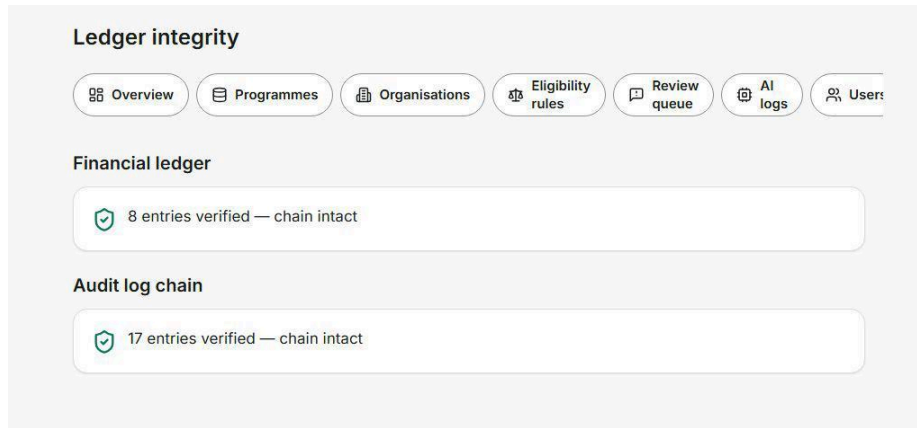


Figure 14. Both chains re-verified in the running prototype: eight financial entries and seventeen audit entries, each checked by recomputing hashes rather than reading a stored flag.

What this proves is bounded, and the code says so in its own comments before this document does. It detects alteration after the fact. It does not prevent an operator with direct database access from rewriting every row and recomputing the whole chain, because there is one writer and no independent second copy to disagree with. It is tamper-evident, not tamper-proof, and calling it a blockchain would be false: there is no distributed consensus and no external witness.

The append itself reads the current head and writes a new row in two separate statements with no transaction. A uniqueness constraint on the previous-hash column means a concurrent second writer fails loudly on insert rather than silently forking the chain, which is the right failure direction, but the losing write leaves its allocation recorded and unanchored. Ordering also depends on millisecond-precision timestamps, so two entries written in the same millisecond have ambiguous order. Issue G-04, and a blocker for the multi-instance deployment in Section 2.

The verification is independently runnable. It is a plain script requiring no server and no staff login, and it calls the same functions the administrator dashboard calls rather than a second implementation that could drift from what the application actually enforces.

```
$ npm run ledger:audit
```

7. Security and privacy

This section states what is enforced, then lists what is not.

7.1 Controls that are implemented

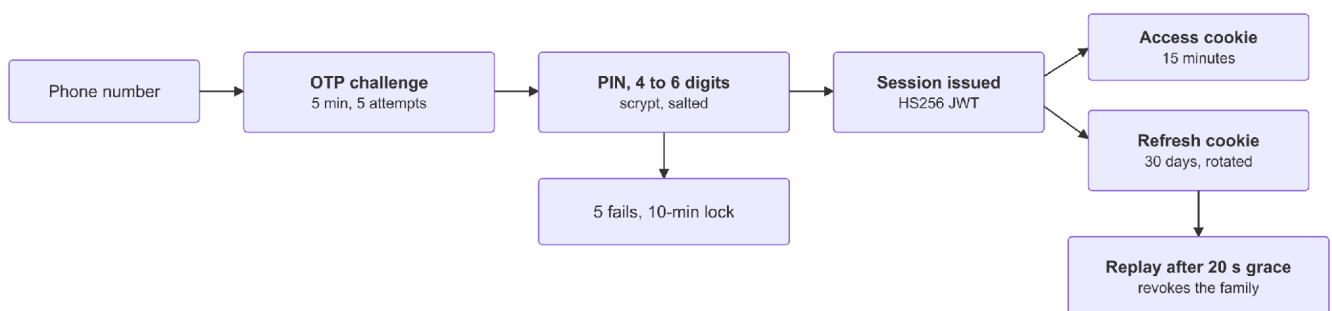


Figure 15. Authentication and session lifetimes, as the code enforces them. Three weaknesses in this flow: token-only guards read the role from the access token and never reload the user, so a demotion can lag by up to the 15-minute

access lifetime (S-06); the one-time-code comparison is plain string equality rather than the constant-time comparison already used for PINs (S-08); and the demo shortcut signs in as a seeded, verified beneficiary account without touching the OTP path (S-11).

Control	Implementation
Credential hashing	scrypt, N=32768 r=8 p=1, 64-byte key, fresh 16-byte random salt per credential, constant-time comparison. Argon2id was specified originally and rejected because it needs native bindings the Workers runtime cannot load; the deviation is documented in code.
Session tokens	HS256 JWT, 15-minute access and 30-day refresh, both httpOnly and secure in production
Refresh rotation	Opaque 48-byte second factor, rotated on use; replay outside a 20-second race window revokes every session for that user and writes an audit event
Brute-force limits	OTP 5-minute validity, 5 attempts, 30-second resend cooldown. PIN 5 attempts then a 10-minute lock; trivial and repeated-digit PINs rejected
Rate limiting	DB-backed token bucket: 120 requests per minute default, 20 for AI, 10 for auth, 30 for voice
Authorisation	Role hierarchy of five roles, plus place-scoped civic authority that is deliberately not rank-based, so a national administrator is not automatically a union officer
Sensitive fields	Medical conditions encrypted at rest with AES-256-GCM and a fresh 12-byte IV per value; never sent to any model
Audit trail	Every staff mutation appended to a hash-chained log with actor, role, before and after
Browser hardening	CSP with default-src self and frame-ancestors none, nosniff, X-Frame-Options DENY, strict referrer policy, scoped permissions policy

7.2 Security issues found, with severity and fix

ID	Issue	Severity	Specified fix
S-03	OTP echo defaults to on in the schema, returning the one-time code in the API response. wrangler.jsonc now sets OTP_DEV_ECHO to false, so the deployed Worker is covered, but the schema default is unchanged and S-02 still has no caller to enforce it.	Critical	Flip the schema default to false and call the S-02 check at boot, so no later deployment can regress silently. The wrangler.jsonc override is a mitigation, not a fix.
S-05	Rate limiting covers 11 of 58 API routes. The token refresh and session endpoints are unprotected, and the bucket update is read-then-write, so concurrent requests can exceed the quota.	High	Apply the default bucket globally in middleware; make the bucket update atomic.
S-06	A suspended or demoted staff member's existing access token remains valid on routes that read the role from the token rather than the database, for up to its 15-minute lifetime.	Medium	Re-check user status on privileged handlers, as the refresh path already does.
S-07	The public transparency page publishes per-union allocation, flag, escalation, and per-programme beneficiary counts with no authentication. With three active beneficiaries in a small union, aggregate plus local knowledge can identify individuals.	High	Suppress cells below a minimum count and round beneficiary totals into bands.
S-08	OTP comparison is a plain string equality, not constant-time.	Low	Use the constant-time comparison already used for PINs.
S-09	No HSTS header, and script-src permits unsafe-inline in production.	Medium	Add HSTS; move to nonce-based script policy.

ID	Issue	Severity	Specified fix
S-10	Flagging is Sybil-vulnerable: the escalation threshold rests on identity verification that is simulated, so a small coordinated group can force or bury escalations.	Medium	Treat as blocked on real identity verification; until then, cap flags per account and log flag provenance.
S-11	The login screen carries a one-click demo shortcut, with a phone number and PIN hardcoded in the client. It signs in as a seeded account that the seed script marks a verified beneficiary in Rangpur, so any visitor can hold a verified civic identity, the credential the escalation denominator (G-01) and the Sybil argument (S-10) both rest on. The button says Rahima Khatun; the seeded account is Rahima Begum.	Medium	Put the shortcut behind a build-time flag so it cannot reach a real deployment, and strip the demo account of verified residency and the right to flag. Reconcile the two names. Until then, treat any flag or issue raised from it as unattributed.

Verify your identity

Reporting problems and voting requires verifying your National ID and your union.

i **AccessAI**
 Verification is simulated in this prototype — the number's format is checked, not matched against a government database.

National ID

 **Simulated verification**

National ID number

Figure 16. Identity verification tells the citizen, on the screen itself, that the check is simulated and the number is only format-checked. The honesty is in the product, not only in this document.

AccessAI
 Transparency

বাংলা English

Open to the public — no sign-in required. Published in keeping with the Union Parishads Act 2009's accountability commitments.

 Ledger intact — no tampering detected
 8 entries verified

Unions

Kaligonj Union
 Chairman: মোঃ জাসিম উদ্দিন
 Allocations
 ₳75,00,000 (4)

Rangpur Sadar, rangpur
 1
 1

Mominpur Union
 Rangpur Sadar, rangpur

Figure 17. The public transparency page, reachable with no login. Issue S-07 concerns exactly this surface: it is the right instinct for accountability and needs small-cell suppression before real beneficiary data appears on it.

8. Reliability and failure modes

8.1 External dependency behaviour

Dependency	Timeout	Retry	Failure behaviour
LLM provider	30 s per attempt, 45 s route ceiling	1, 400 ms backoff	Degrades to the deterministic composer; citizen sees a degraded badge
Speech to text	45 s	None	Fails closed; microphone disabled with a stated reason
Text to speech	NONE (issue X-01)	None	A hung provider is bounded only by the platform
Overpass / OpenStreetMap	30 s	None, though failures are classified retryable	Seeded locations still served; map marked unavailable
Map tiles	8 s	None	504 and a grey grid
SMS providers	NONE (issue X-01)	None	Throws; OTP delivery fails
Vision moderation	NONE (issue X-01)	None	Marked unavailable and routed to human review, which is the safe direction

9. Operations

ID	Operational gap	Severity	Detail and fix
O-01	Background jobs never run on their own	High	Six jobs exist, including staleness detection and data-retention enforcement, but they are triggered only by a staff member clicking in the admin panel and execute inline in that HTTP request. A retention job that is never clicked never runs. Add a scheduled trigger, move execution off the request path, add retry and a dead-letter record.
O-02	No backup or restore of any kind	Critical	There is no snapshot, export, or restore path in the repository. A tamper-evident financial ledger with no durability guarantee protects integrity while risking total loss. Add automated snapshots with a stated recovery objective and a rehearsed restore.
O-03	Schema changes are pushed, not migrated	High	Deployment uses a schema-diff push with no ordered migration history and no rollback. One generated migration file exists and is applied by nothing. Adopt versioned migrations before any deployment holding real data.
O-04	A destructive script is one word away from a routine one	Medium	The seed script deletes roughly twenty tables including the financial ledger, and the documented reset script is missing entirely, so the command in package.json fails. Remove the broken entry and require an explicit confirmation flag for destructive scripts.
O-05	One job iterates all citizens inside a single request	Medium	Scheduled notifications loop over every citizen in one handler, bounded only by platform limits. Batch it.
O-06	Observability is a dashboard, not telemetry	Medium	Request IDs, AI logs with model and prompt version, grounding-failure counts, and job history all exist and are visible in the admin panel, but nothing is exported to a metrics or alerting system, so no one is paged when something breaks.

10. Validation: what is measured, and how to reproduce it

Component	Method	Result	Command
Whole system	Automated test suite	789 tests across 26 files, all passing, 7.17 s	npm test
Eligibility engine	Unit tests: three-valued logic, all 13 comparators, group combinators, weighted scoring, replayable trace	28 tests passing	npm test
Retrieval relevance	20 labelled known-item queries, 10 Bangla and 10 English	Recall@1 85%, Recall@3 90%, MRR 0.867. Bangla 8/10 against English 10/10	npx tsx scripts/retrieval-eval.ts
Ledger and audit chains	Independent hash recomputation outside the application	Both chains verify; breaks report the offending row	npm run ledger:audit
Confidence scoring	Tests asserting the score can never exceed its verification ceiling	16 tests passing	npm test
Accessibility contrast	WCAG ratios computed from the actual design tokens	27 tests passing	npm run test:a11y
Anomaly detectors	Unit tests for all five detectors	14 tests passing	npm test
Grounded generation	Zero-citation flag counted in logs	Metric only, no pass or fail gate (issue A-02)	Admin AI logs
Speech accuracy	Not measured	No model adapted yet; floors defined in Section 5.5	Not applicable

12. Access channels and accessibility

A product for citizens with limited literacy and shared low-specification phones cannot assume a smartphone. Three channels exist.

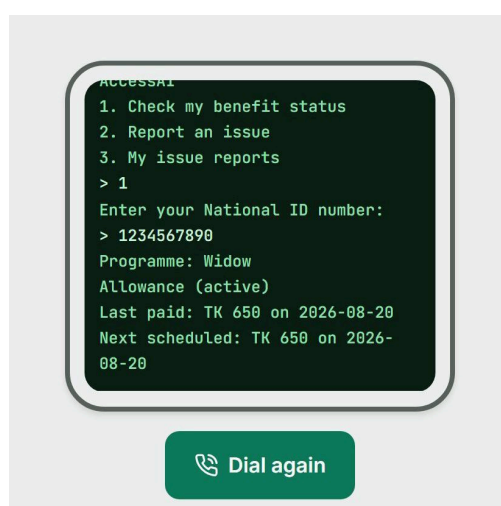


Figure 18. The USSD channel: benefit status and issue reporting from any handset, over a menu identical to what a telecom gateway would render, calling the same server logic as the web app.

Voice covers navigation, dictation, and read-aloud. Command routing is deterministic and works offline; server transcription requires a key and is disabled with a stated reason when absent. The Bangla vocabulary hint sent to the speech provider is a curated list of 26 domain phrases covering

programme names, navigation words, and money terms, with a shorter command-mode variant capped at 560 characters to stay inside the model's prompt window. The team measured prompt bleed during development, where a silent recording returned a word from the hint list, and documented it rather than ignoring it.

Accessibility is verified rather than asserted: 27 tests compute real WCAG contrast ratios from the design tokens, including a documented erratum where a design specification value fell below its own stated floor.

The login screen carries a one-click demo shortcut that skips phone and OTP registration. It signs in as a seeded account marked a verified beneficiary in Rangpur, with the phone number and PIN hardcoded. Because that account carries verified civic standing, it is recorded as issue S-11 at medium severity. The deployment is also reachable without cloning the repository, by scanning the QR code at public/shebar-janala-live-qr.png.

13. Conclusion

The eligibility engine, the retrieval pipeline, the governance layer, and both hash chains are real, tested, and runnable by anyone who clones the repository. The measurement in Section 5 is the first honest quality number this project has produced about its own AI, and it found a language gap in the direction that matters most. The register in Section 13 lists 37 things that are wrong or missing, ordered by how much they could hurt a citizen.

For a widow in Rangpur, none of that is abstract. She is not helped by an eligibility engine that is elegant, nor harmed by a hash chain that is only tamper-evident. She is helped when the amount on the screen is one a government office will honour, and when the flag she raises about her union's money reaches somebody who has to answer for it.